

DAY 1

Introduction to SQL

Learn how to logically communicate with and command a database.

✓ What is SQL

✓ Basic Concepts

✓ Basic Commands

Real-Life Example: School Register

Think of data storage like physical items:

 Notebook → Database

 Page → Table

 Row → Record

 Column → Field

ROLL NO	NAME	CLASS	MARKS
1	Rahul	5th	80
2	Anita	5th	90

What is SQL?

Structured Query Language

It is the standard language used to communicate with relational databases.

💬 Human language → Talk to people

🗄️ SQL → Talk to database

Example request: "Show all students."

```
SELECT * FROM students;
```



Core Concepts

TERM	TECHNICAL MEANING	REAL LIFE EXAMPLE
Database	Collection of interconnected tables	 School Office
Table	Structured grid of rows & columns	 Student Register
Row	One single complete record	 One Student's Data
Column	One specific detail type	 Name / Marks
Query	Actionable instruction	 Teacher's Command

The Core Four: Basic Commands

We will master these foundational instructions step by step to manipulate any database.



SELECT

Used to **read** or view existing data from the database without altering it.



INSERT

Used to **add** entirely new rows of data into a specific table.



UPDATE

Used to **change** or modify data that already exists within a table.



DELETE

Used to **remove** specific rows of data from a table permanently.

Hour 2

Practical Start

Step 1: Create Database

Before creating tables, you must establish the main container (database).

```
CREATE DATABASE school;
```

Select the database to start using it:

```
USE school;
```



Step 2 – Create Table

You are building the structure of a digital school register.

```
CREATE TABLE students ( id INT, name  
VARCHAR(50), class VARCHAR(20), marks INT  
);
```

- Defines the column names.
- Assigns Data Types (**INT** for numbers, **VARCHAR** for text).



Step 3 – Insert Data

</> The Query

```
INSERT INTO students VALUES (1,  
'Rahul', '5th', 80), (2, 'Anita',  
'5th', 90), (3, 'Rohit', '6th', 75);
```



Data Stored Successfully

The table **students** is no longer empty. It now permanently holds the 3 rows of records we just executed.

Step 4 – View Data



Fetching Records

This command retrieves and shows all stored records from the table.

```
SELECT * FROM students;
```

💡 The asterisk (*) is a wildcard that means "ALL columns".

Practice Questions (Easy)

- 1 Create a database named `shop`.
- 2 Create a table named `products` with columns:
(id, name, price)
- 3 Insert **3 products** into the table.
- 4 Write a query to show all products.

Practice Questions (Medium)

Create a table named `employees` ensuring you use correct data types:

Required Columns:

- ✓ `id`
- ✓ `name`
- ✓ `salary`
- ✓ `department`



Final Step

Write the query to
Insert 4 employees.



Why store data in tables instead of physical notebooks?

CORE ADVANTAGES

 Easy Search  Fast Updates  Secure Storage  No Physical Damage

Mini Real-Life Task

Imagine you own a mobile shop.

1. Create Table: `mobiles`

Columns Required:

- id
- price
- brand
- stock

2. Populate Data

Insert at least 5 mobiles into your inventory.



What You **Learned** Today

After Day 1, you have the foundational skills to:

-  Understand core database concepts
-  Create a new database (**CREATE DATABASE**)
-  Create structured tables (**CREATE TABLE**)
-  Insert new data records (**INSERT INTO**)
-  View stored data (**SELECT**)



You are now a beginner Data Manager!

PREVIEW

Tomorrow: Day 2

We will dive deep into the **SELECT** statement and learn advanced filtering.

```
SELECT * FROM students  
WHERE marks > 80;
```

You will start thinking like a true data analyst. 🔥

Image Sources



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